

Numerical TESTs for Partial Differential Equations (PDEs)

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1 Advection equation

1.1 Governing Equation

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\mathbf{u}\phi) = 0.$$

Please write down its semi-discretized form based on finite volume method and develop numerical solver based on explicit schemes for both temporal and spatial discretizations.

1.2 Numerical Example

1.2.1 Advection equation on a sine wave test

The accuracy of linear advection equation is first evaluated by using a sine function [1] given as

$$\phi_0(x, y) = \sin 2\pi(x + y)$$

and transported with a constant and uniform advection velocity $(u, v) = (1, 1)$.

A square domain $[0, 1]^2$ is partitioned by both triangular and quadrilateral elements with periodic boundary condition imposed in $x - y$ directions. The computation is carried out on gradually refined grids with the time step adjusted by $\text{CFL} = 0.2$ throughout all computations.

Please present the L_1 errors and convergence rates at $t = 1.0$.

1.2.2 Solid rotation of a complex profile

To assess the capability of a scheme to resolve discontinuities, we compute solid rotation of a complex profile including a slotted disk, a cone and a smooth hump as shown in [2, 3]. The rotated velocity is given as $\mathbf{u} = (0.5 - y, x - 0.5)$ on a square domain of $[0, 1]^2$ which is partitioned by triangular and quadrilateral elements with 100 vertices along each edge.

Please plot contour profiles after one round revolution at time $t = 2\pi$.

References

- [1] C. Hu and C.-W. Shu. Weighted essentially non-oscillatory schemes on triangular meshes. *Journal of Computational Physics*, 150:97–127, 1999.
- [2] R. J. LeVeque. High-resolution conservative algorithms for advection in incompressible flow. *SIAM Journal on Numerical Analysis*, 33:627–665, 1996.

- [3] J. S. Park, S. H. Yoon, and C. Kim. Multi-dimensional limiting process for hyperbolic conservation laws on unstructured grids. *Journal of Computational Physics*, 257:788–812, 2010.